

Quantum Algorithm-2

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Math on Deutsch- Jozsa algorithm

Book: Quantum Computing Explained

Algorithm Pg: 207

Maths: Pg: 208

Consider a function with two inputs such that $f(x)=1$. Explicitly show that the Deutsch- Jozsa algorithm works in this case by generating the vector $|y\rangle = |00\rangle$ as the final output.

The initial input state will be $|\psi_{in}\rangle = |0\rangle|0\rangle|1\rangle$. Applying Hadamard gates to this state gives

$$|\psi'\rangle = \frac{1}{2\sqrt{2}}(|000\rangle - |001\rangle + |010\rangle - |011\rangle + |100\rangle - |101\rangle + |110\rangle - |111\rangle)$$

Next we apply U_f to the system, obtaining

$$U_f|x, y\rangle = |x, y \oplus f(x)\rangle$$

$$|\psi''\rangle = \frac{1}{2\sqrt{2}}(|001\rangle - |000\rangle + |011\rangle - |010\rangle + |101\rangle - |100\rangle + |111\rangle - |110\rangle)$$

The final step is to apply $H^{\otimes 2}$ to the first two qubits. This gives

$$\begin{aligned} |\psi_{out}\rangle = & \frac{1}{2\sqrt{2}} \left[\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) |1\rangle - \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) |0\rangle \right. \\ & + \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) |1\rangle - \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) |0\rangle \\ & + \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) |1\rangle - \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) |0\rangle \\ & \left. + \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) |1\rangle - \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) |0\rangle \right] \end{aligned}$$

Expanding out all the terms gives

$$\begin{aligned}
 |\psi_{out}\rangle = \frac{1}{4\sqrt{2}}[& (|00\rangle + |01\rangle + |10\rangle + |11\rangle)|1\rangle - (|00\rangle + |01\rangle + |10\rangle + |11\rangle)|0\rangle \\
 & + (|00\rangle - |01\rangle + |10\rangle - |11\rangle)|1\rangle - (|00\rangle - |01\rangle + |10\rangle - |11\rangle)|0\rangle \\
 & + (|00\rangle + |01\rangle - |10\rangle - |11\rangle)|1\rangle - (|00\rangle + |01\rangle - |10\rangle - |11\rangle)|0\rangle \\
 & + (|00\rangle - |01\rangle - |10\rangle + |11\rangle)|1\rangle - (|00\rangle - |01\rangle - |10\rangle + |11\rangle)|0\rangle]
 \end{aligned}$$

Now we want to factor these terms to put them in the form with the third qubit as $(|0\rangle - |1\rangle/\sqrt{2})$. We get

$$\begin{aligned}
 |\psi_{out}\rangle = \frac{1}{4} \left[& -(|00\rangle + |01\rangle + |10\rangle + |11\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \right. \\
 & - (|00\rangle - |01\rangle + |10\rangle - |11\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\
 & - (|00\rangle + |01\rangle - |10\rangle - |11\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\
 & \left. - (|00\rangle - |01\rangle - |10\rangle + |11\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \right]
 \end{aligned}$$

Hence

$$|\psi_{out}\rangle = -|00\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

Measurement on the first two qubits gives 00, confirming that this is a constant function.

Suppose that $f(00)=f(01)=0$, $f(10)=f(11)=1$. Apply the Deutsch-Josza algorithm and show that at least one of the first two qubits ends up as a 1.

The input state is again $|\psi_{in}\rangle = |0\rangle|0\rangle|1\rangle$, and the application of the Hadamard gates gives

$$|\psi'\rangle = \frac{1}{2\sqrt{2}}(|000\rangle - |001\rangle + |010\rangle - |011\rangle + |100\rangle - |101\rangle + |110\rangle - |111\rangle)$$

We apply U_f to the system, obtaining

$$U_f|x, y\rangle = |x, y \oplus f(x)\rangle$$

$$|\psi''\rangle = \frac{1}{2\sqrt{2}}(|000\rangle - |001\rangle + |010\rangle + |011\rangle + |101\rangle - |100\rangle + |111\rangle - |110\rangle)$$

Applying Hadamard gates to the first two inputs gives

$$\begin{aligned} |\psi_{out}\rangle = & \frac{1}{4} \left[\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) |0\rangle - \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) |1\rangle \right. \\ & + \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) |0\rangle + \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) |1\rangle \\ & + \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) |1\rangle - \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) |0\rangle \\ & \left. + \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) |1\rangle - \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) |0\rangle \right] \end{aligned}$$

Again, we expand this result to obtain

$$\begin{aligned}
 |\psi_{out}\rangle = \frac{1}{4\sqrt{2}}[&(|00\rangle + |01\rangle - |10\rangle - |11\rangle)|1\rangle + (|00\rangle + |01\rangle + |10\rangle + |11\rangle)|0\rangle \\
 &- (|00\rangle + |01\rangle + |10\rangle + |11\rangle)|1\rangle + (|00\rangle - |01\rangle + |10\rangle - |11\rangle)|0\rangle \\
 &+ (|00\rangle - |01\rangle + |10\rangle - |11\rangle)|1\rangle - (|00\rangle + |01\rangle - |10\rangle - |11\rangle)|0\rangle \\
 &+ (|00\rangle - |01\rangle - |10\rangle + |11\rangle)|1\rangle - (|00\rangle - |01\rangle - |10\rangle + |11\rangle)|0\rangle]
 \end{aligned}$$

We rearrange as follows:

$$\begin{aligned}
 |\psi_{out}\rangle = \frac{1}{4} \Bigg[&(|00\rangle + |01\rangle + |10\rangle + |11\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\
 &+ (|00\rangle - |01\rangle + |10\rangle - |11\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\
 &- (|00\rangle + |01\rangle - |10\rangle - |11\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\
 &- \left(|00\rangle - |01\rangle - |10\rangle + |11\rangle \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \Bigg]
 \end{aligned}$$

So we find that

$$|\psi_{out}\rangle = |10\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

Measurement on the first two qubits returns 10, since at least one qubit was 1. This shows that the function is balanced.

Grover's Algorithm

Book: Quantum Computing Explained

Algorithm Pg: 218

Book: Quantum Computing for Computer Scientists

Pg: 195

Grover's Search Algorithm

- **Problem:** Given a function $f(x)$ mapping n bits to 1 bit $f(x): \{0,1\}^n \rightarrow \{0,1\}$, determine the value of x such that $f(x) = 1$. We have a *promise* that the value is unique.
- Classically we need to evaluate $O(N)$ values, $N = 2^n$.



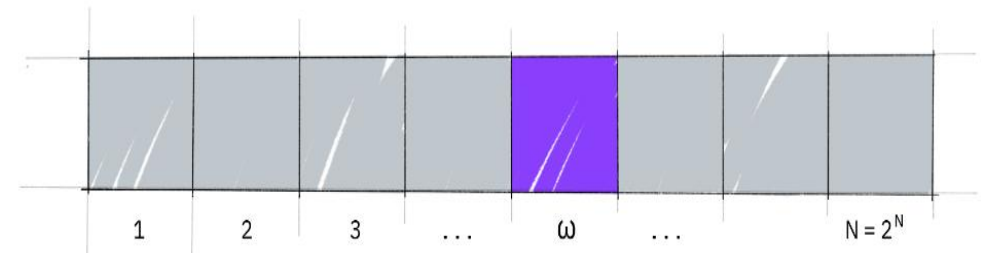
Lov Grover

1996

- Grover's algorithm enables a search with $O(\sqrt{N})$ queries.

- Quantum algorithm has two crucial steps:

1. Calculate the function on all values simultaneously.
2. Reflect about the equal superposition state.



- These steps are repeated $O(\sqrt{N})$ times.

Preparing the Search Space

The first step of Grover's algorithm is the initial state preparation. the search space is all possible values we need to search through to find the answer we want. For example, if we have 3 qubits, our list is the states

$$|000\rangle, |001\rangle, \dots |111\rangle \text{ (i.e the states } |0\rangle \rightarrow |7\rangle).$$

So, in this case the size of our search space will be

$$N = 2^3 = 8.$$

Creating the Oracle

- The second and most important step of Grover's algorithm is the oracle. Oracles add a negative phase to the solution states so they can stand out from the rest and be measured, i.e. for any state $|x\rangle$ in the computational basis:

$$U_{\omega}|x\rangle = \begin{cases} |x\rangle & \text{if } x \neq \omega \\ -|x\rangle & \text{if } x = \omega \end{cases}$$

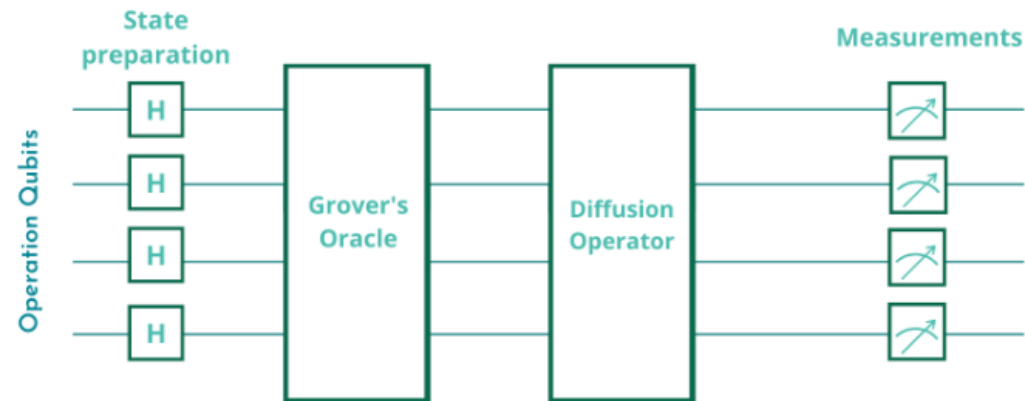
$$U_{\omega}|x\rangle = (-1)^{f(x)}|x\rangle$$

Grover's search algorithm

- We start in the state

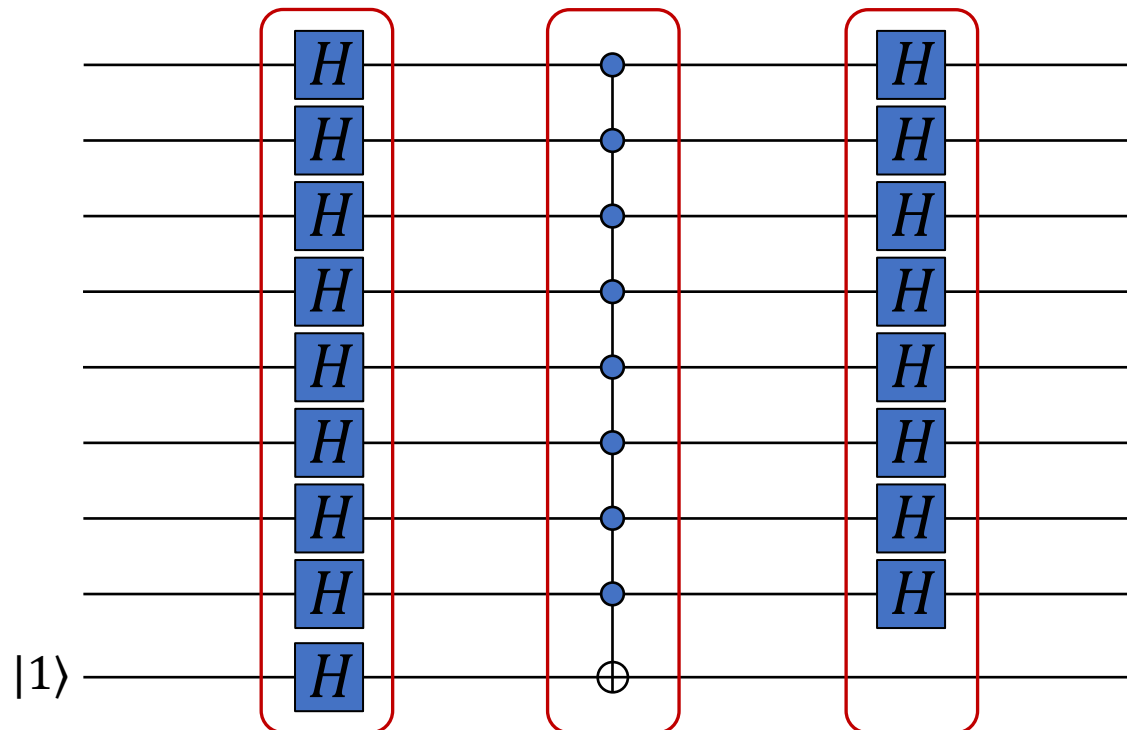
$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{x=1}^N |x\rangle$$

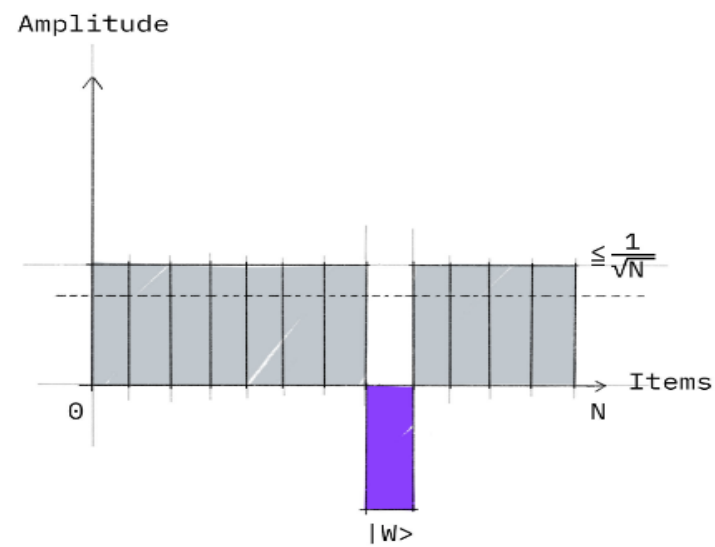
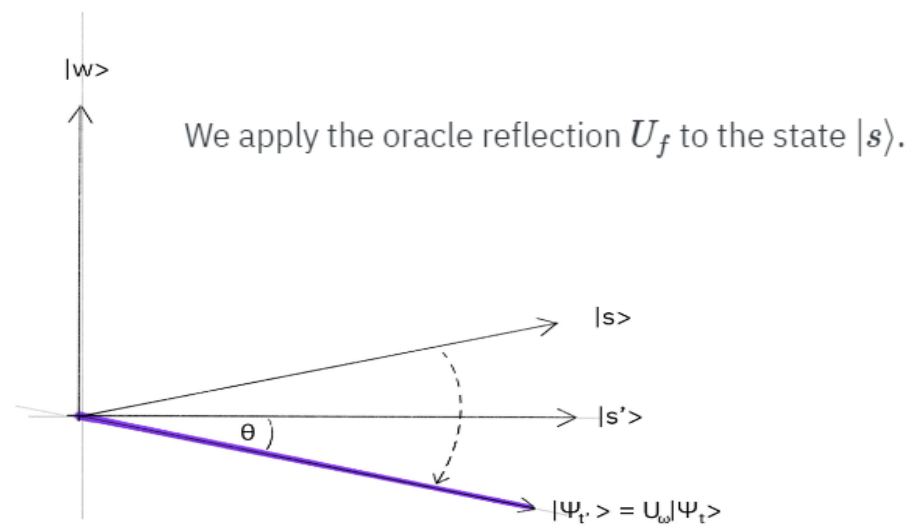
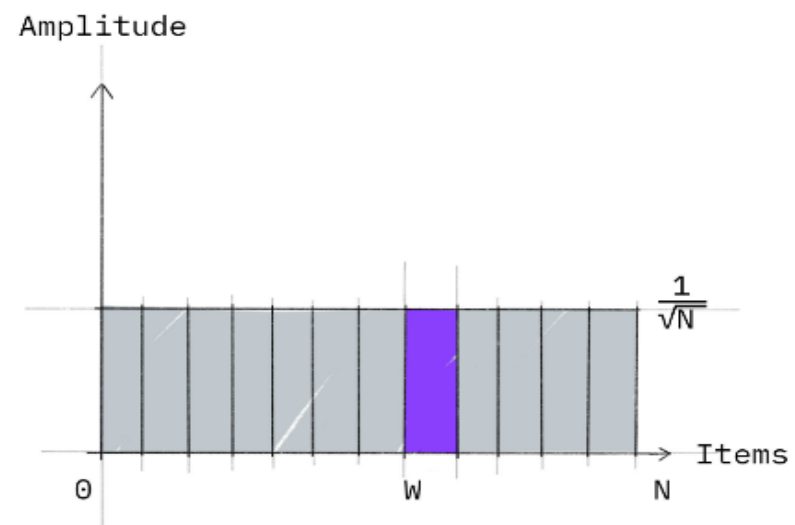
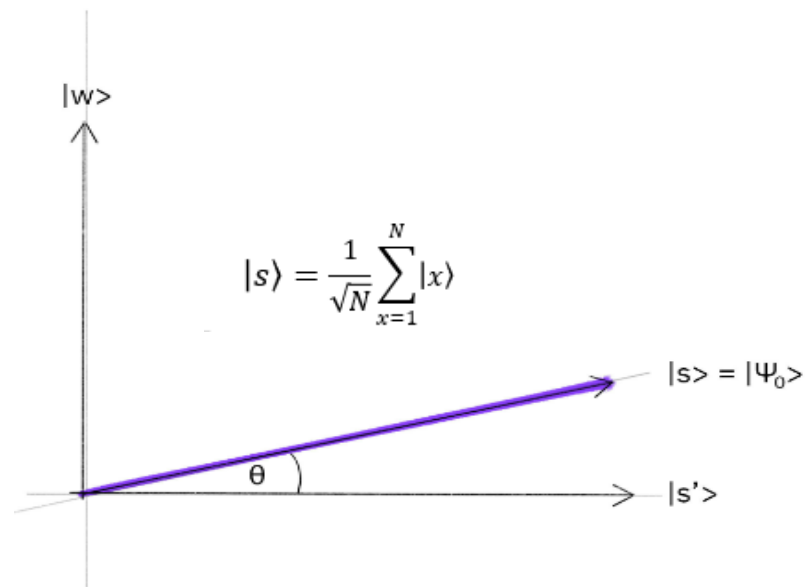
- We repeat the following $\sim \sqrt{N}$ times:
 1. (phase inversion) Apply the function calculation $U_f = \mathbb{I} - 2|\omega\rangle\langle\omega|$
 2. (inversion about mean) Apply the reflection operation $U_s = 2|s\rangle\langle s| - \mathbb{I}$.
- The system will be in the state $|\omega\rangle$ such that $f(\omega) = 1$.
- But how do we perform U_s ?



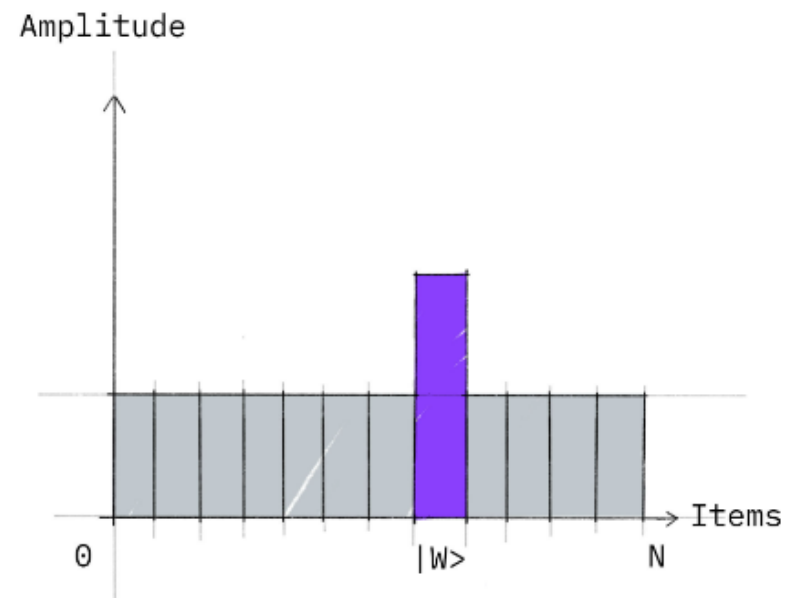
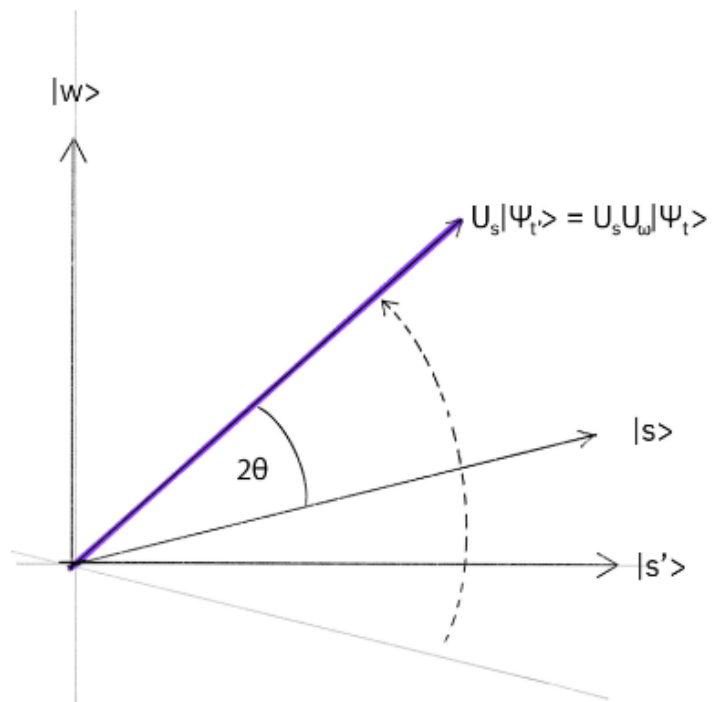
Grover's search algorithm

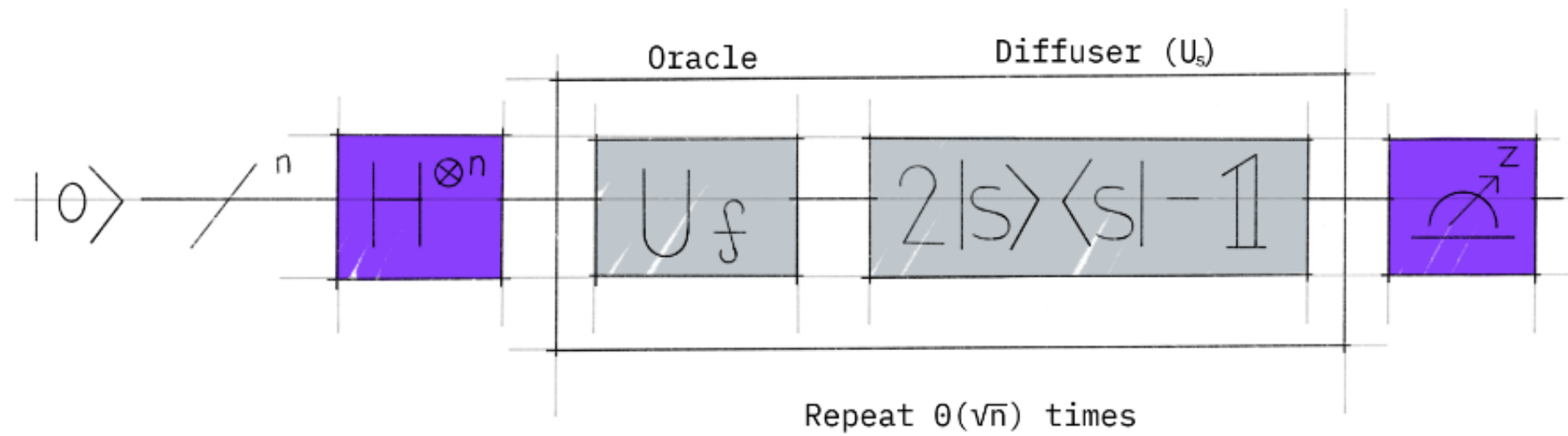
- But how do we perform U_s ?
- First, invert the operation that prepares $|s\rangle$ - just Hadamards on all qubits.
- Then, reflect about $|0\rangle$; i.e. apply $2|0\rangle\langle 0| - \mathbb{I}$.
- Lastly, repeat the operation that prepares $|s\rangle$.





apply an additional reflection (U_s) about the state $|s\rangle$: $U_s = 2|s\rangle\langle s| - \mathbb{1}$.





Grover's search algorithm

- **Another way of looking at it: Rotations**

- Define the ket perpendicular to $|\omega\rangle$.

$$|s'\rangle = \frac{1}{\sqrt{N-1}} \sum_{x \neq \omega} |x\rangle$$

- Then

$$|s\rangle = \cos \phi |s'\rangle + \sin \phi |\omega\rangle$$

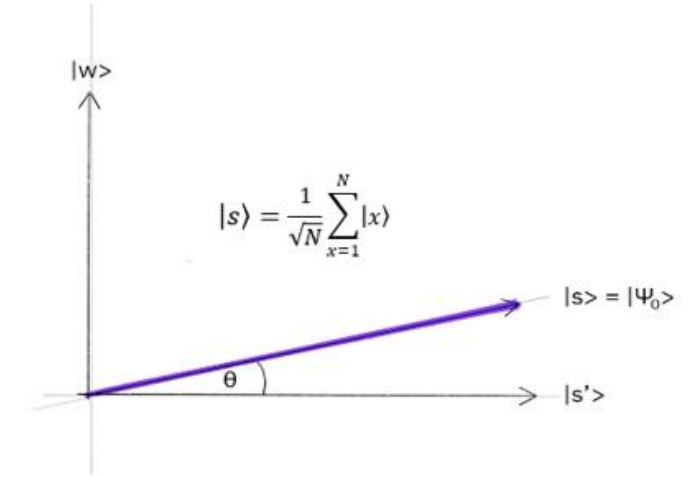
with $\sin \phi = 1/\sqrt{N}$.

- The operation U_f reflects $|\omega\rangle$ and leaves all orthogonal states unchanged.

$$U_f = \mathbb{I} - 2|\omega\rangle\langle\omega|$$

- The action of U_f on $|s'\rangle$ is

$$U_f |s'\rangle = (\mathbb{I} - 2|\omega\rangle\langle\omega|) |s'\rangle = |s'\rangle$$



From now onwards we will denote the angle as ϕ for denoting general purpose

Grover's search algorithm

- **Another way of looking at it: Rotations**

- The action of U_f on $|s'\rangle$ is

$$U_f|s'\rangle = |s'\rangle$$

- The action of U_s on $|\omega\rangle$ is

$$\begin{aligned}U_s|\omega\rangle &= (2|s\rangle\langle s| - \mathbb{I})|\omega\rangle \\&= 2\langle s|\omega\rangle|s\rangle - |\omega\rangle \\&= 2\sin\phi|s\rangle - |\omega\rangle\end{aligned}$$

- Now use

$$|s\rangle = \cos\phi|s'\rangle + \sin\phi|\omega\rangle$$

- This gives

$$\begin{aligned}U_s|\omega\rangle &= 2\sin\phi(\cos\phi|s'\rangle + \sin\phi|\omega\rangle) - |\omega\rangle \\&= 2\sin\phi\cos\phi|s'\rangle - (1 - 2\sin^2\phi)|\omega\rangle \\&= \sin 2\phi|s'\rangle - \cos 2\phi|\omega\rangle\end{aligned}$$

Grover's search algorithm

The action of U_f on $|s'\rangle$ is

$$U_f|s'\rangle = |s'\rangle$$

- The action of U_s on $|\omega\rangle$ is

$$U_s|\omega\rangle = \sin 2\phi |s'\rangle - \cos 2\phi |\omega\rangle$$

- The action of U_s on $|s'\rangle$ is

$$\begin{aligned} U_s|s'\rangle &= (2|s\rangle\langle s| - \mathbb{I})|s'\rangle \\ &= 2\langle s|s'\rangle|s\rangle - |s'\rangle \\ &= 2\cos\phi |s\rangle - |s'\rangle \end{aligned}$$

- Now use

$$|s\rangle = \cos\phi |s'\rangle + \sin\phi |\omega\rangle$$

- This gives

$$\begin{aligned} U_s|s'\rangle &= 2\cos\phi (\cos\phi |s'\rangle + \sin\phi |\omega\rangle) - |s'\rangle \\ &= (2\cos^2\phi - 1)|s'\rangle + 2\sin\phi \cos\phi |\omega\rangle \\ &= \cos 2\phi |s'\rangle + \sin 2\phi |\omega\rangle \end{aligned}$$

Grover's search algorithm

$$\begin{aligned}U_f|s'\rangle &= |s'\rangle \\U_s|\omega\rangle &= \sin 2\phi |s'\rangle - \cos 2\phi |\omega\rangle \\U_s|s'\rangle &= \cos 2\phi |s'\rangle + \sin 2\phi |\omega\rangle\end{aligned}$$

- Consider a state written as

$$|\psi\rangle = \cos \theta |s'\rangle + \sin \theta |\omega\rangle$$

- The action of U_f on $|\psi\rangle$ is

$$U_f|\psi\rangle = \cos \theta |s'\rangle - \sin \theta |\omega\rangle$$

- Then applying U_s gives $U_s U_f |\psi\rangle$

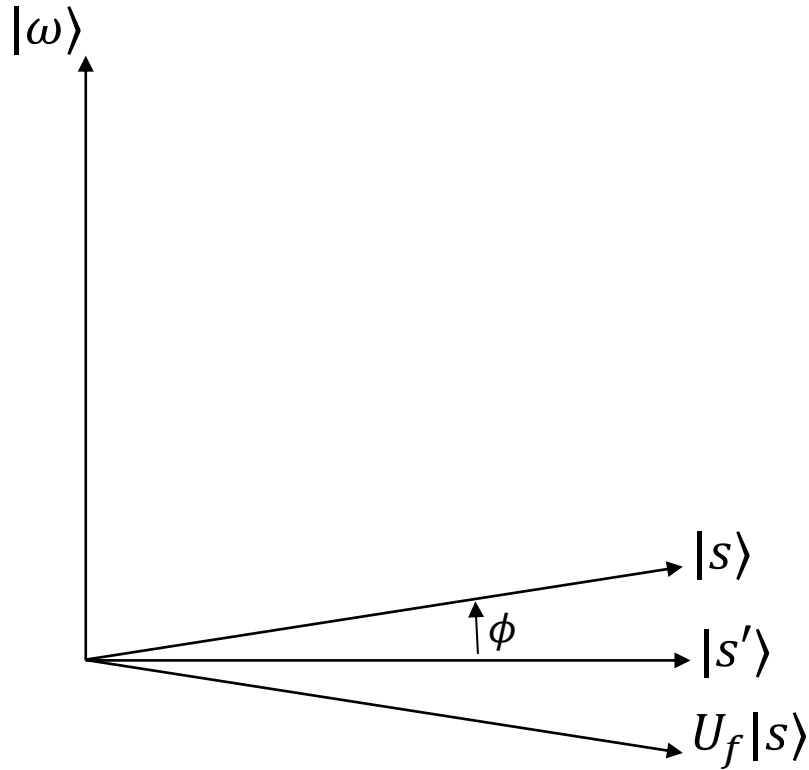
$$\begin{aligned}&= \cos \theta [\cos 2\phi |s'\rangle + \sin 2\phi |\omega\rangle] - \sin \theta [\sin 2\phi |s'\rangle - \cos 2\phi |\omega\rangle] \\&= [\cos \theta \cos 2\phi - \sin \theta \sin 2\phi] |s'\rangle + [\cos \theta \sin 2\phi + \sin \theta \cos 2\phi] |\omega\rangle \\&= \cos(\theta + 2\phi) |s'\rangle + \sin(\theta + 2\phi) |\omega\rangle\end{aligned}$$

Grover's search algorithm

$$\begin{aligned}U_f|s'\rangle &= |s'\rangle \\U_s|\omega\rangle &= \sin 2\phi |s'\rangle - \cos 2\phi |\omega\rangle \\U_s|s'\rangle &= \cos 2\phi |s'\rangle + \sin 2\phi |\omega\rangle\end{aligned}$$

- The Grover iteration gives

$$\begin{aligned}|\psi\rangle &= \cos \theta |s'\rangle + \sin \theta |\omega\rangle \\U_s U_f |\psi\rangle &= \cos(\theta + 2\phi) |s'\rangle + \sin(\theta + 2\phi) |\omega\rangle\end{aligned}$$



Grover's search algorithm

$$U_f|s'\rangle = |s'\rangle$$

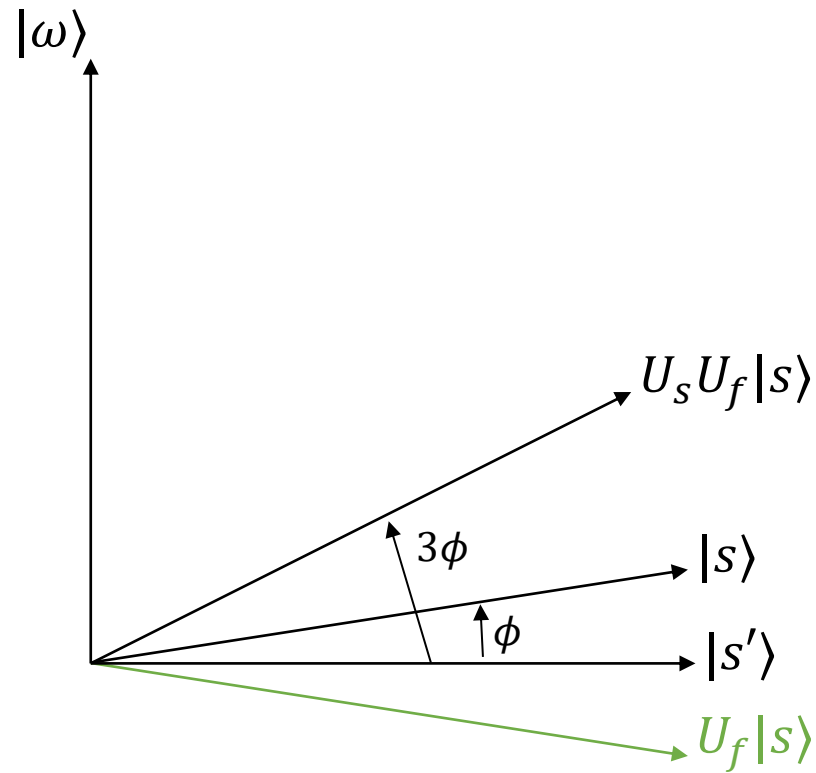
$$U_s|\omega\rangle = \sin 2\phi |s'\rangle - \cos 2\phi |\omega\rangle$$

$$U_s|s'\rangle = \cos 2\phi |s'\rangle + \sin 2\phi |\omega\rangle$$

- The Grover iteration gives

$$|\psi\rangle = \cos \theta |s'\rangle + \sin \theta |\omega\rangle$$

$$U_s U_f |\psi\rangle = \cos(\theta + 2\phi) |s'\rangle + \sin(\theta + 2\phi) |\omega\rangle$$



Grover's search algorithm

$$U_f|s'\rangle = |s'\rangle$$

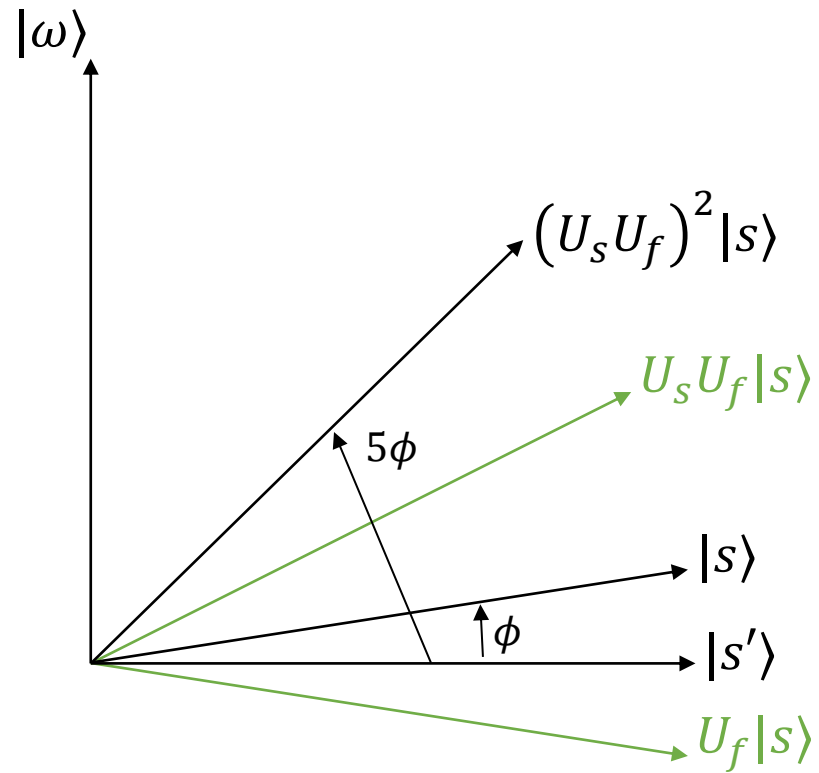
$$U_s|\omega\rangle = \sin 2\phi |s'\rangle - \cos 2\phi |\omega\rangle$$

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Grover's search algorithm

$$U_f|s'\rangle = |s'\rangle$$

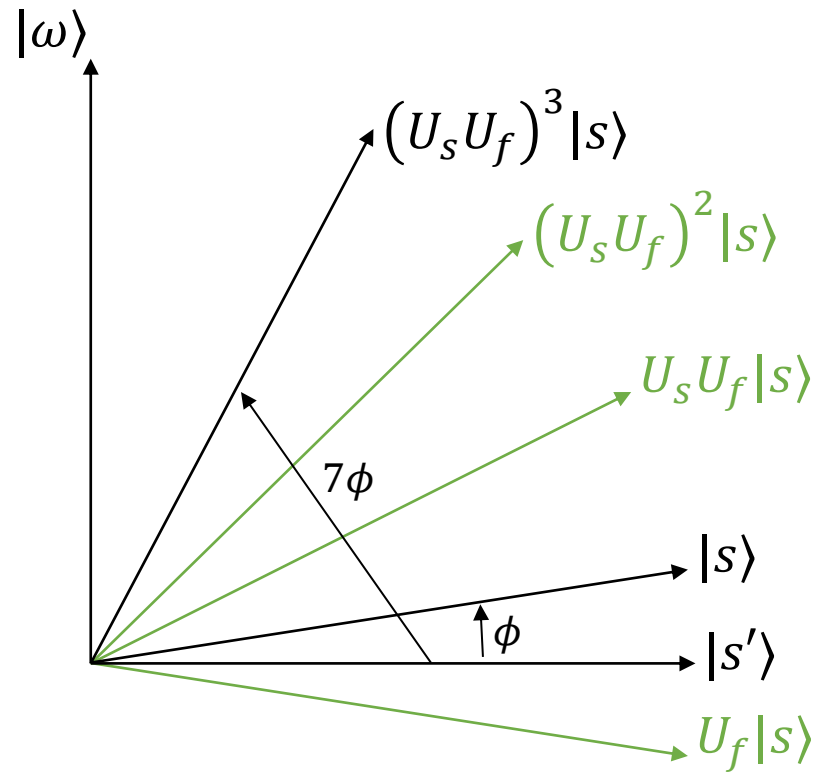
$$U_s|\omega\rangle = \sin 2\phi |s'\rangle - \cos 2\phi |\omega\rangle$$

$$U_s|s'\rangle = \cos 2\phi |s'\rangle + \sin 2\phi |\omega\rangle$$

- The Grover iteration gives

$$|\psi\rangle = \cos \theta |s'\rangle + \sin \theta |\omega\rangle$$

$$U_s U_f |\psi\rangle = \cos(\theta + 2\phi) |s'\rangle + \sin(\theta + 2\phi) |\omega\rangle$$



Grover's search algorithm

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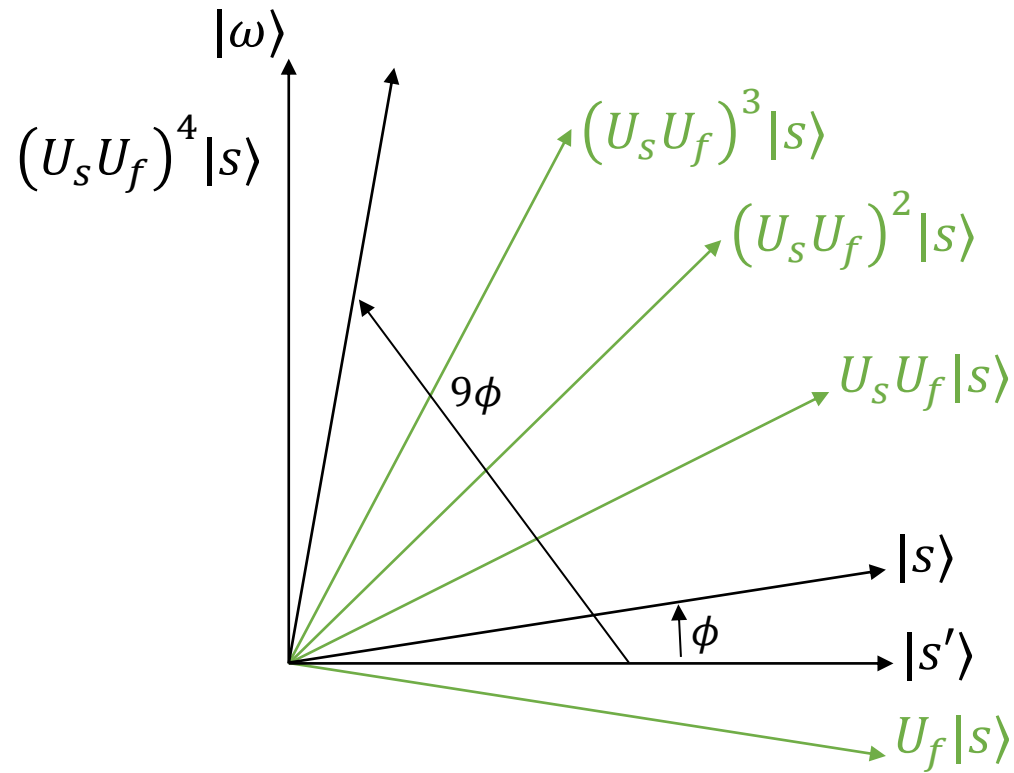
$$U_s|\omega\rangle = \sin 2\phi |s'\rangle - \cos 2\phi |\omega\rangle$$

$$U_s|s'\rangle = \cos 2\phi |s'\rangle + \sin 2\phi |\omega\rangle$$

- The Grover iteration gives

$$|\psi\rangle = \cos \theta |s'\rangle + \sin \theta |\omega\rangle$$

$$U_s U_f |\psi\rangle = \cos(\theta + 2\phi) |s'\rangle + \sin(\theta + 2\phi) |\omega\rangle$$



Grover's search algorithm

$$U_f|s'\rangle = |s'\rangle$$

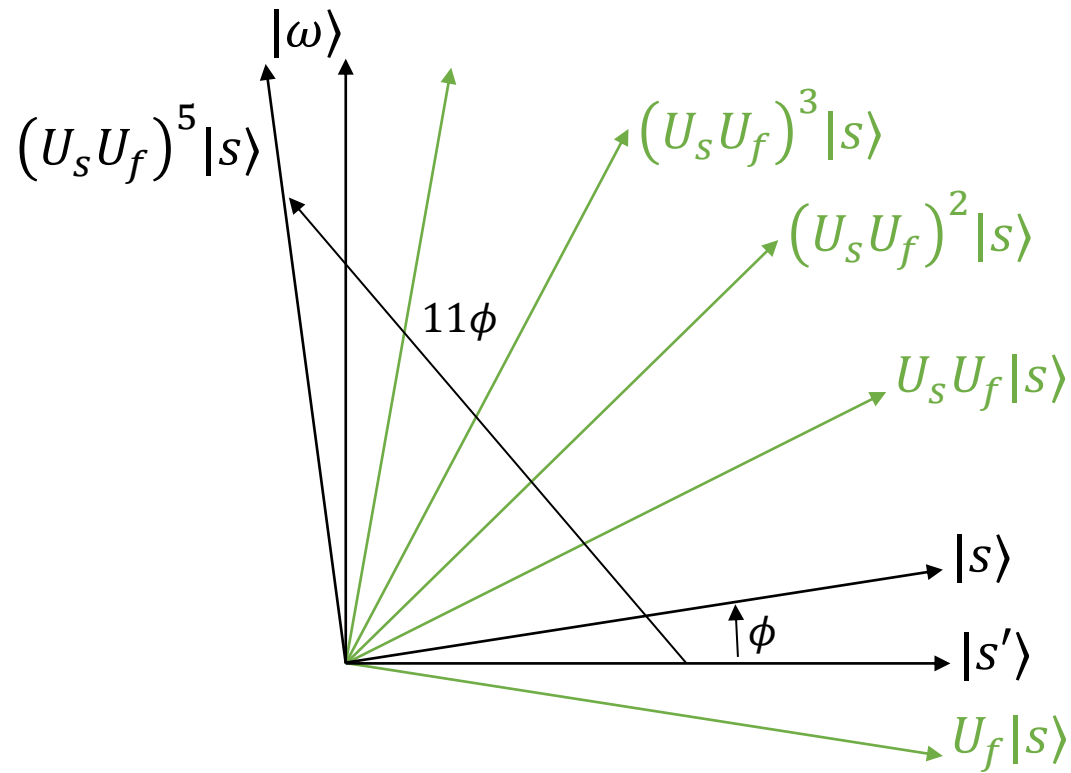
$$U_s|\omega\rangle = \sin 2\phi |s'\rangle - \cos 2\phi |\omega\rangle$$

$$U_s|s'\rangle = \cos 2\phi |s'\rangle + \sin 2\phi |\omega\rangle$$

- The Grover iteration gives

$$|\psi\rangle = \cos \theta |s'\rangle + \sin \theta |\omega\rangle$$

$$U_s U_f |\psi\rangle = \cos(\theta + 2\phi) |s'\rangle + \sin(\theta + 2\phi) |\omega\rangle$$



Grover's search algorithm

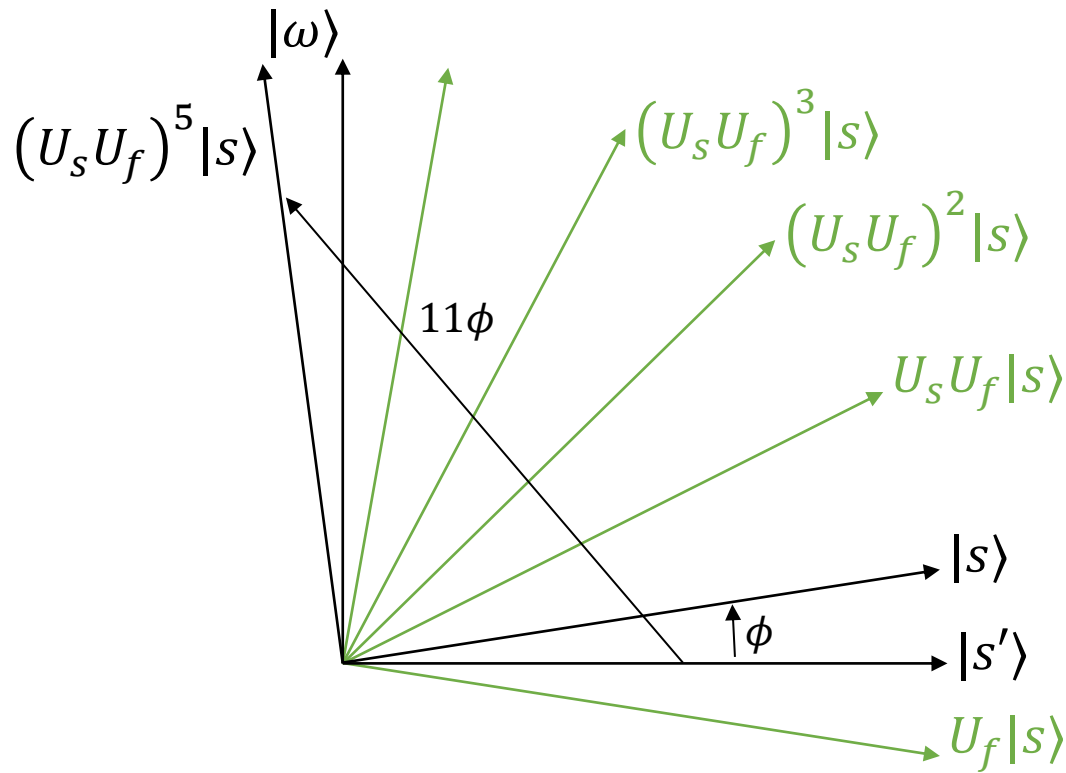
$$\sin \phi = 1/\sqrt{N}.$$

- In general

$$(U_s U_f)^k |s\rangle = \cos[(2k+1)\phi] |s'\rangle + \sin[(2k+1)\phi] |\omega\rangle$$

- For solution we want

$$(2k+1)\phi \approx \pi/2 \quad \Rightarrow \quad (2k+1) \arcsin(1/\sqrt{N}) \approx \pi/2$$



$$k_{\text{opt}} \approx \frac{\pi}{4} \sqrt{N} - \frac{1}{2}$$

For $N = 42$

$$k_{\text{opt}} \approx 4.58996 \dots$$

Grover's search algorithm

- **What about multiple targets?**
- In general we have m values of x such that $f(x) = 1$.
- Let us define

$$|\omega\rangle = \frac{1}{\sqrt{m}} \sum_{n=1}^m |\omega_n\rangle$$

where ω_n are the values of x such that $f(x) = 1$.

- Define a modified $|s'\rangle$

$$|s'\rangle = \frac{1}{\sqrt{N-m}} \sum_{x \notin \{\omega_n\}} |x\rangle$$

- Then

$$|s\rangle = \cos \phi |s'\rangle + \sin \phi |\omega\rangle$$

with $\sin \phi = \sqrt{m/N}$.

- The analysis is the same, except with the modified value of ϕ .

Define projector

$$P_\omega = \sum_{n=1}^m |\omega_n\rangle\langle\omega_n|$$

and reflector

$$U_f = \mathbb{I} - 2P_\omega$$

$$k_{\text{opt}} \approx \frac{\pi}{4} \sqrt{N} \quad \text{becomes} \quad k_{\text{opt}} \approx \frac{\pi}{4} \sqrt{\frac{N}{m}}$$

Algorithm: Quantum search

Inputs: (1) a black box oracle O which performs the transformation $O|x\rangle|q\rangle = |x\rangle|q \oplus f(x)\rangle$, where $f(x) = 0$ for all $0 \leq x < 2^n$ except x_0 , for which $f(x_0) = 1$; (2) $n + 1$ qubits in the state $|0\rangle$.

Outputs: x_0 .

Runtime: $O(\sqrt{2^n})$ operations. Succeeds with probability $O(1)$.

Procedure:

1. $|0\rangle^{\otimes n}|0\rangle$ initial state
2. $\rightarrow \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right]$ apply $H^{\otimes n}$ to the first n qubits, and HX to the last qubit
3. $\rightarrow \left[(2|\psi\rangle\langle\psi| - I)O \right]^R \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right]$ apply the Grover iteration $R \approx \lceil \pi\sqrt{2^n}/4 \rceil$ times.
 $\approx |x_0\rangle \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right]$
4. $\rightarrow x_0$ measure the first n qubits

Proof: $(2|s\rangle\langle s| - I)|\psi\rangle \Rightarrow v_i \mapsto 2a - v_i$

Let the system have N basis states: $|\psi\rangle = \sum_{i=1}^N v_i |i\rangle$
 where v_i are amplitudes.

Equal superposition state $|s\rangle = \frac{1}{\sqrt{N}} \sum_{j=1}^N |j\rangle$

$$\langle s|\psi\rangle = \frac{1}{\sqrt{N}} \sum_{j=1}^N v_j$$

Define the average: $a = \frac{1}{N} \sum_{j=1}^N v_j$

$$\langle s|\psi\rangle = \sqrt{N} a$$

$$|s\rangle\langle s|\psi\rangle = |s\rangle(\langle s|\psi\rangle) = |s\rangle(\sqrt{N}a)$$

Substitute $|s\rangle$:
$$= \frac{1}{\sqrt{N}} \sum_{i=1}^N |i\rangle \cdot \sqrt{N}a$$

$$= \sum_{i=1}^N a|i\rangle$$

$2|s\rangle\langle s||\psi\rangle = \sum_{i=1}^N 2a|i\rangle$ Multiply by 2

$$(2|s\rangle\langle s| - I)|\psi\rangle$$

$$= \sum_{i=1}^N 2a|i\rangle - \sum_{i=1}^N v_i|i\rangle$$

$$= \sum_{i=1}^N (2a - v_i)|i\rangle$$

$v_i \longrightarrow 2a - v_i$

We will use this in our next math.

Example 6.4.2 Let us look at an example of an execution of this algorithm. Let f be a function that picks out the string “101.” The states after each step will be

$$|\varphi_1\rangle = \begin{bmatrix} 000 & 001 & 010 & 011 & 100 & 101 & 110 & 111 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^T, \quad (6.135)$$

$$|\varphi_2\rangle = \begin{bmatrix} 000 & 001 & 010 & 011 & 100 & 101 & 110 & 111 \\ \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} \end{bmatrix}^T, \quad (6.136)$$

$$|\varphi_{3a}\rangle = \begin{bmatrix} 000 & 001 & 010 & 011 & 100 & 101 & 110 & 111 \\ \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} \end{bmatrix}^T. \quad (6.137)$$

The average of these numbers is

$$a = \frac{7 * \frac{1}{\sqrt{8}} - \frac{1}{\sqrt{8}}}{8} = \frac{\frac{6}{\sqrt{8}}}{8} = \frac{3}{4\sqrt{8}}. \quad (6.138)$$

Calculating the inversion about the mean we have

$$-v + 2a = -\frac{1}{\sqrt{8}} + \left(2 \times \frac{3}{4\sqrt{8}}\right) = \frac{1}{2\sqrt{8}} \quad (6.139)$$

and

$$-v + 2a = \frac{1}{\sqrt{8}} + \left(2 \times \frac{3}{4\sqrt{8}}\right) = \frac{5}{2\sqrt{8}}. \quad (6.140)$$

Book: Quantum Computing
for Computer Scientists
pg no: 202-204

All amplitudes equal

This is the uniform state $|s\rangle$

One amplitude becomes **negative**

$$U_f : |w\rangle \rightarrow -|w\rangle$$

average amplitude

For marked state

Formula from last slide

Thus, we have

$$|\varphi_{3b}\rangle = \begin{bmatrix} \frac{000}{2\sqrt{8}}, & \frac{001}{2\sqrt{8}}, & \frac{010}{2\sqrt{8}}, & \frac{011}{2\sqrt{8}}, & \frac{100}{2\sqrt{8}}, & \frac{101}{2\sqrt{8}}, & \frac{110}{2\sqrt{8}}, & \frac{111}{2\sqrt{8}} \end{bmatrix}^T \quad (6.141)$$

A phase inversion will give us

$$|\varphi_{3a}\rangle = \begin{bmatrix} \frac{000}{2\sqrt{8}}, & \frac{001}{2\sqrt{8}}, & \frac{010}{2\sqrt{8}}, & \frac{011}{2\sqrt{8}}, & \frac{100}{2\sqrt{8}}, & -\frac{101}{2\sqrt{8}}, & \frac{110}{2\sqrt{8}}, & \frac{111}{2\sqrt{8}} \end{bmatrix}^T \quad (6.142)$$

The average of these numbers is

$$a = \frac{7 * \frac{1}{2\sqrt{8}} - \frac{5}{2\sqrt{8}}}{8} = \frac{1}{8\sqrt{8}}. \quad (6.143)$$

Calculating for the inversion about the mean we have

$$-v + 2a = -\frac{1}{2\sqrt{8}} + \left(2 \times \frac{1}{8\sqrt{8}}\right) = -\frac{1}{4\sqrt{8}} \quad (6.144)$$

and

$$-v + 2a = \frac{5}{2\sqrt{8}} + \left(2 \times \frac{1}{8\sqrt{8}}\right) = \frac{11}{4\sqrt{8}}. \quad (6.145)$$

Hence, we have

$$|\varphi_{3b}\rangle = \begin{bmatrix} \frac{000}{4\sqrt{8}}, & \frac{001}{4\sqrt{8}}, & \frac{010}{4\sqrt{8}}, & \frac{011}{4\sqrt{8}}, & \frac{100}{4\sqrt{8}}, & \frac{101}{4\sqrt{8}}, & \frac{110}{4\sqrt{8}}, & \frac{111}{4\sqrt{8}} \end{bmatrix}^T \quad (6.146)$$

For the record, $\frac{11}{4\sqrt{8}} = 0.97227$ and $\frac{-1}{4\sqrt{8}} = -0.08839$. Squaring the numbers gives us the probability of measuring those numbers. When we measure the state in Step 4, we will most likely get the state

$$|\varphi_4\rangle = \begin{bmatrix} \frac{000}{4\sqrt{8}}, & \frac{001}{4\sqrt{8}}, & \frac{010}{4\sqrt{8}}, & \frac{011}{4\sqrt{8}}, & \frac{100}{4\sqrt{8}}, & \frac{101}{4\sqrt{8}}, & \frac{110}{4\sqrt{8}}, & \frac{111}{4\sqrt{8}} \end{bmatrix}^T, \quad (6.147)$$

which is exactly what we wanted. $\square$

- Another way to represent Grover's Algorithm
- Pg: 248: Book: Quantum Computation and Quantum Information, Michael A. Nielsen

Reference

Original Paper:

A fast quantum mechanical algorithm for database search: Lov K. Grover

Link: <https://arxiv.org/pdf/quant-ph/9605043.pdf>

YouTube link:

https://www.youtube.com/watch?v=m4zs_fqS0Zw